

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 11: Areas Related to Circles

Step-by-step solutions to the textbook exercises: 14 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 11.1

- Q1** Find the area of a sector of a circle of radius 6 cm with sector angle 60° . (Use $\pi = \frac{22}{7}$ unless told otherwise.)

SOLUTION

$$\text{Area of a sector} = \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 6 \times 6 = \frac{132}{7}.$$

Answer: $\frac{132}{7} \text{ cm}^2$ (about 18.86 cm²)

- Q2** Find the area of a quadrant of a circle whose circumference is 22 cm.

SOLUTION

$$2\pi r = 22 \text{ gives } r = \frac{22 \times 7}{2 \times 22} = \frac{7}{2} \text{ cm.}$$

$$\text{Area of a quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{77}{8}.$$

Answer: $\frac{77}{8} \text{ cm}^2$

- Q3** The minute hand of a clock is 14 cm long. Find the area it sweeps in 5 minutes.

SOLUTION

In 60 minutes the hand turns through 360° , so in 5 minutes it turns through 30° .

$$\text{Area} = \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{154}{3}.$$

Answer: $\frac{154}{3} \text{ cm}^2$ (about 51.33 cm²)

- Q4** A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of (i) the minor segment (ii) the major sector. (Use $\pi = 3.14$.)

SOLUTION

Area of the minor sector = $\frac{90}{360} \times 3.14 \times 100 = 78.5 \text{ cm}^2$. Area of the right triangle formed by the radii = $\frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$.

(i) Minor segment = $78.5 - 50 = 28.5 \text{ cm}^2$.

(ii) Major sector = area of the circle – minor sector = $314 - 78.5 = 235.5 \text{ cm}^2$.

Answer: (i) 28.5 cm^2 (ii) 235.5 cm^2

- Q5** In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find (i) the length of the arc (ii) the area of the sector (iii) the area of the segment formed by the corresponding chord.

SOLUTION

(i) Arc length = $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22 \text{ cm}$.

(ii) Area of the sector = $\frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231 \text{ cm}^2$.

(iii) The triangle formed by the two radii and the chord is equilateral (two equal sides with an angle of 60° between them). Its area is $\frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4} \text{ cm}^2$. Segment = $231 - \frac{441\sqrt{3}}{4} \text{ cm}^2$.

Answer: (i) 22 cm (ii) 231 cm^2 (iii) $\left(231 - \frac{441\sqrt{3}}{4}\right) \text{ cm}^2$

- Q6** A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the minor and major segments. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.)

SOLUTION

Area of the sector = $\frac{60}{360} \times 3.14 \times 225 = 117.75 \text{ cm}^2$.

The triangle is equilateral with side 15 cm: area = $\frac{\sqrt{3}}{4} \times 225 = 1.73 \times 56.25 = 97.3125 \text{ cm}^2$.

Minor segment = $117.75 - 97.3125 = 20.4375 \text{ cm}^2$. Major segment = $3.14 \times 225 - 20.4375 = 706.5 - 20.4375 = 686.0625 \text{ cm}^2$.

Answer: minor segment 20.4375 cm^2 , major segment 686.0625 cm^2

- Q7** A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding segment. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.)

SOLUTION

$$\text{Area of the sector} = \frac{120}{360} \times 3.14 \times 144 = 150.72 \text{ cm}^2.$$

Draw the perpendicular from the centre to the chord. It makes two right triangles with angles of 60° at the centre: the height is $12 \cos 60^\circ = 6$ cm and half the chord is $12 \sin 60^\circ = 6\sqrt{3}$ cm.

$$\text{Area of the triangle} = \frac{1}{2} \times 12\sqrt{3} \times 6 = 36\sqrt{3} = 62.28 \text{ cm}^2.$$

$$\text{Answer: segment} = 150.72 - 62.28 = 88.44 \text{ cm}^2$$

- Q8** A horse is tied to a peg at one corner of a square field of side 15 m by a rope 5 m long. Find (i) the area of the field the horse can graze (ii) the increase in the grazing area if the rope were 10 m long. (Use $\pi = 3.14$.)

SOLUTION

The corner angle is 90° , so the horse grazes a quadrant of a circle whose radius is the length of the rope.

$$(i) \frac{1}{4} \times 3.14 \times 5^2 = 19.625 \text{ m}^2.$$

$$(ii) \text{ With a 10 m rope: } \frac{1}{4} \times 3.14 \times 10^2 = 78.5 \text{ m}^2. \text{ Increase} = 78.5 - 19.625 = 58.875 \text{ m}^2.$$

$$\text{Answer: (i) } 19.625 \text{ m}^2 \text{ (ii) } 58.875 \text{ m}^2$$

- Q9** A brooch is made of silver wire as a circle of diameter 35 mm, with 5 diameters that divide it into 10 equal sectors. Find (i) the total length of wire needed (ii) the area of each sector.

SOLUTION

$$(i) \text{ Circumference} = \frac{22}{7} \times 35 = 110 \text{ mm. Five diameters} = 5 \times 35 = 175 \text{ mm. Total} = 285 \text{ mm.}$$

$$(ii) \text{ Each sector has angle } 36^\circ: \text{ area} = \frac{1}{10} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = \frac{385}{4} \text{ mm}^2.$$

$$\text{Answer: (i) } 285 \text{ mm (ii) } \frac{385}{4} \text{ mm}^2 \text{ (96.25 mm}^2\text{)}$$

- Q10** An umbrella has 8 equally spaced ribs. Taking it as a flat circle of radius 45 cm, find the area between two consecutive ribs.

SOLUTION

The ribs divide the circle into 8 equal sectors, each with angle 45° .

$$\text{Area} = \frac{1}{8} \times \frac{22}{7} \times 45 \times 45 = \frac{22275}{28}.$$

$$\text{Answer: } \frac{22275}{28} \text{ cm}^2 \text{ (about } 795.54 \text{ cm}^2\text{)}$$

Q11 A car has two wipers that do not overlap. Each blade is 25 cm long and sweeps through 115° . Find the total area cleaned at each sweep.

SOLUTION

$$\text{Area swept by one blade} = \frac{115}{360} \times \frac{22}{7} \times 25 \times 25.$$

$$\text{Total for two blades} = 2 \times \frac{115}{360} \times \frac{22}{7} \times 625 = \frac{158125}{126}.$$

Answer: $\frac{158125}{126} \text{ cm}^2$ (about 1254.96 cm^2)

Q12 A lighthouse spreads a red light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which ships are warned. (Use $\pi = 3.14$.)

SOLUTION

$$\text{Area} = \frac{80}{360} \times 3.14 \times 16.5 \times 16.5 = \frac{2}{9} \times 3.14 \times 272.25 = 189.97.$$

Answer: about 189.97 km^2

Q13 A round table cover of radius 28 cm has six equal designs, each the segment between a side of a regular hexagon drawn in the circle and the arc above it. Find the cost of making the designs at ₹0.35 per cm^2 . (Use $\sqrt{3} = 1.7$.)

SOLUTION

Each side of the hexagon subtends 60° at the centre, so each design is a segment of angle 60° , and its triangle is equilateral with side 28 cm.

$$\text{One sector} = \frac{60}{360} \times \frac{22}{7} \times 28 \times 28 = \frac{1232}{3} \text{ cm}^2. \text{ One triangle} = \frac{\sqrt{3}}{4} \times 28^2 = 1.7 \times 196 = 333.2 \text{ cm}^2.$$

$$\text{Six designs} = 6 \left(\frac{1232}{3} - 333.2 \right) = 2464 - 1999.2 = 464.8 \text{ cm}^2.$$

$$\text{Cost} = 464.8 \times 0.35 = ₹162.68.$$

Answer: ₹162.68

Q14 The area of a sector of angle p (in degrees) of a circle of radius R is (A) $\frac{p}{180} \times 2\pi R$ (B) $\frac{p}{180} \times \pi R^2$ (C) $\frac{p}{360} \times 2\pi R$ (D) $\frac{p}{720} \times 2\pi R^2$.

SOLUTION

$$\text{Area of a sector} = \frac{p}{360} \times \pi R^2 = \frac{p}{720} \times 2\pi R^2.$$

Answer: (D)

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